MAHATMA GANDHI UNIVERSITY KOTTAYAM

MGU-BVOC (HONOURS) REGULAR EXAMINATION NOVEMBER 2025

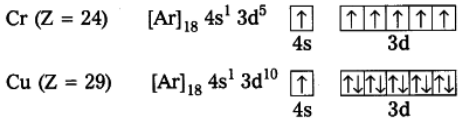
FIRST SEMSTER

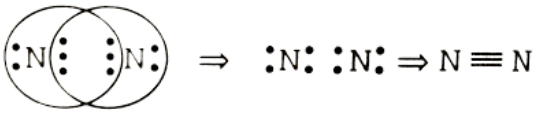
MPC- MG1MPCCHY100- BASIC THEORETICAL AND ANALYTICAL CHEMISTRY

(2025 ADMISSION ONWARDS)

DURATION: 1 HOUR 30 MINUTES MAXIMUM MARKS:50

PART A

1. $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^4$
2. 

3. The de Broglie equation states that all moving particles have an associated wavelength, given by the formula $\lambda=h/p$, where λ is the wavelength, h is Planck's constant, and $p$is the momentum of the particle. This relationship can also be expressed as λ=h/(mv), where m is the mass and v is the velocity, and it establishes the wave-particle duality of matter.
4. Aufbau principle, Hund's rule & Pauli's exclusion principle.
5. The fine structure of the hydrogen spectrum is the splitting of individual spectral lines into multiple, closely spaced lines.
6. The principal quantum number, denoted by n, describes the energy level and size of an electron's shell in an atom. It can be any positive integer (n=1,2, 3,). A higher value of 'n' indicates a higher energy level, a larger orbital size, and a greater distance from the nucleus. For example, (n=1) represents the first energy shell (K shell), while (n=2) represents the second (L shell).
7. Spherical shape
8. 

9. Increases across a period.
10. Atomic radius is a measure of the size of an atom, defined as the distance from the nucleus to the outermost electron shell.
11. Any example give full mark.
12. Acid: A proton (H+) donor. Base: A proton (H+) acceptor.

PART B

13. The magnetic quantum number (ml) describes the spatial orientation of an atomic orbital, while the spin quantum number (ms) describes the intrinsic angular momentum of an electron, which is its "spin". The magnetic quantum number has values from (-l) to (+l), and the spin quantum number has two possible values: (+1/2) or (-1/2), corresponding to spin-up and spin-down states.

Feature	Magnetic Quantum Number (m_l)	Spin Quantum Number (m_s)
What it describes	The orientation of an orbital in space.	The intrinsic angular momentum of an electron, or its "spin".
Possible values	Any integer from $-l$ to $+l$, where l is the azimuthal quantum number.	Only $+1/2$ or $-1/2$.
Relationship	Determines how many orbitals are available within a subshell and their orientation.	Specifies the two possible spin states of an electron, often called "spin-up" ($+1/2$) and "spin-down" ($-1/2$).
Example	For a p-subshell where $l = 1$, the magnetic quantum number can be $-1, 0$, or $+1$, corresponding to the three different orientations of the p orbitals (p_x, p_y, p_z).	For any electron, the spin quantum number will be either $+1/2$ or $-1/2$.

14. The principal quantum number, denoted by n, describes the energy level and size of an electron's shell in an atom. It can be any positive integer (n=1,2,3…). A higher value of 'n' indicates a higher energy level, a larger orbital size, and a greater distance from the nucleus. For example, (n=1) represents the first energy shell (K shell), while (n=2) represents the second (L shell). Energy Level: It determines the principal energy level of an electron. A higher 'n' means higher energy. Shell Size: As 'n' increases, the size of the electron shell also increases. Number of Orbitals: The number of orbitals in each shell is equal to (n^2). The maximum number of electrons a shell can hold is given by the formula ($2n^2$).
15. Low ionization energy: The metal atom should have a low ionization energy, meaning it requires little energy to lose its outermost electron(s) and form a positive ion (cation). High electron gain enthalpy: The non-metal atom should have a high electron gain enthalpy (or a high electron affinity), which is the energy change when an electron is added to form a negative ion (anion). This means the non-metal readily accepts an electron.
    High lattice energy: The resulting ionic compound should have a high lattice energy. Lattice energy is the energy released when a crystal lattice is formed from gaseous ions, and a high value indicates a very stable compound with strong ionic bonds.
16. .

The molality (m) of a solution is defined as the moles of solute divided by the mass of the solvent in kilograms:

$$m = \frac{\text{moles of solute}}{\text{mass of solvent (kg)}}$$

Substituting the provided values for the moles of NaCl and the mass of the solvent into the equation:

$$m = \frac{0.150\,\text{mol}}{0.75\,\text{kg}}$$

The calculation yields the molality value:

$$m = 0.2\,\text{mol/kg}$$

17. Simple distillation and fractional distillation are both separation techniques that work by exploiting differences in the boiling points of liquids in a mixture. The key difference

is that fractional distillation is used for liquids with similar boiling points, while simple distillation is for liquids with widely different boiling points.

Simple Distillation

Simple distillation involves a single cycle of vaporization and condensation. It is effective for:

Separating a liquid from a non-volatile solute (e.g., obtaining pure water from salt water).

Separating two miscible liquids if their boiling points differ by at least 25°C to 70°C (or more, depending on purity requirements).

Fractional Distillation

Fractional distillation is a more advanced and efficient method used to separate components with similar or close boiling points (a difference of less than 25°C). It uses a special piece of equipment called a fractionating column, which is packed with materials like glass beads or rings to increase the surface area.

This column allows for multiple, repeated cycles of vaporization and condensation (known as theoretical plates) as the vapor rises.

Components with lower boiling points rise higher in the column before condensing and being collected, while those with higher boiling points condense lower down and flow back into the flask.

18. Primary standards are highly pure, stable, and non-hygroscopic substances used to calibrate secondary standards, while secondary standards are less pure, more reactive, and are standardized against a primary standard for routine analytical use. Primary standards, like pure sodium carbonate, are used to determine unknown concentrations accurately, whereas secondary standards, such as a standardized sodium hydroxide solution, are used for everyday analyses like instrument calibration.

    Primary standards Definition: Extremely pure substances with known properties used to make a standard solution. Properties: High purity (often (99.9%). Highly stable and not easily affected by air or moisture. Anhydrous (no water molecules attached). High molecular weight, making them easier to weigh accurately. Low reactivity or not reactive at all.

    Examples: Sodium carbonate, potassium hydrogen phthalate, and pure metals.

    Definition: Less pure substances or solutions that are standardized against a primary standard. Properties: Less pure than primary standards. Less stable and potentially reactive. Can be hygroscopic (absorb moisture from the air). Examples: Commercially available sodium hydroxide (NaOH) and hydrochloric acid (HCl) solutions.

PART C

19. (a) An orbit is a fixed, two-dimensional, circular path that electrons were historically thought to follow around the nucleus (based on the Bohr model). This concept is largely outdated in modern atomic theory.

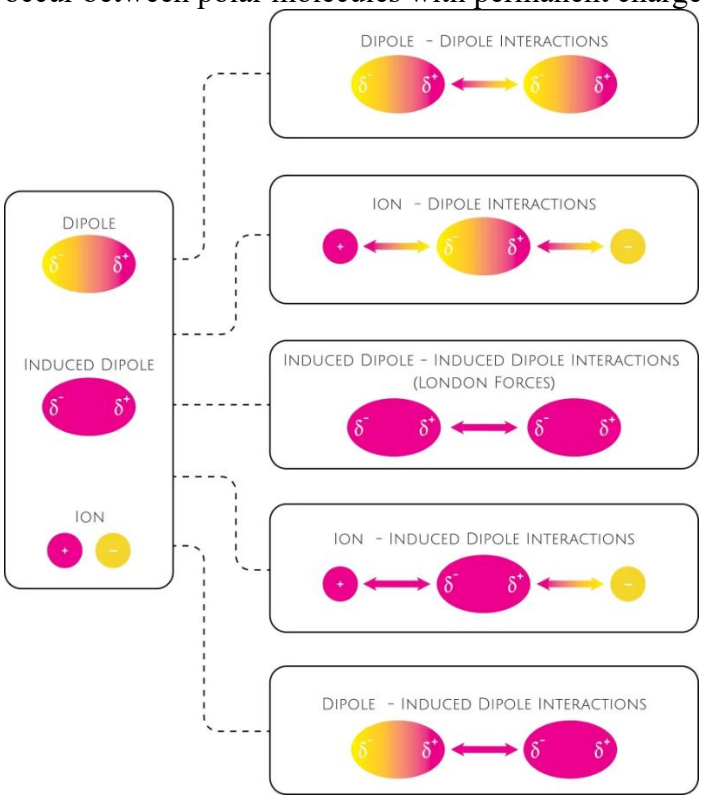
    An orbital is a three-dimensional region of space around the nucleus where there is the highest probability of finding an electron (based on the modern Quantum Mechanical Model).

    Draw the shapes of orbit and orbitals.

    (b) The Heisenberg Uncertainty Principle states that it's impossible to know both the exact position and the exact momentum (velocity) of a subatomic particle simultaneously.

- **Mathematical expression**: The relationship is often expressed as an inequality: $\Delta x \cdot \Delta p \geq \frac{\hbar}{2}$, where Δx is the uncertainty in position, Δp is the uncertainty in momentum, and $\hbar$ is the reduced Planck constant.

20. (a) H-bonding, or hydrogen bonding, is a weak but important intermolecular force that occurs when a hydrogen atom covalently bonded to a highly electronegative atom (like oxygen, nitrogen, or fluorine) is attracted to another nearby electronegative atom.
    Types of hydrogen bonds
    Intermolecular hydrogen bonding: The hydrogen bond forms between two different molecules. A classic example is the hydrogen bonding between separate water (H2O) molecules.
    Intramolecular hydrogen bonding: The hydrogen bond forms between two atoms within the same molecule.
    (b) Van der Waals forces are weak intermolecular forces that arise from temporary or permanent fluctuations in electron distribution within molecules, causing transient or permanent partial charges. These forces are weaker than covalent or ionic bonds and can be separated into two main types: London dispersion forces, which occur in all molecules due to random electron movement, and dipole-dipole interactions, which occur between polar molecules with permanent charges.



21. Ionization enthalpy is the minimum energy required to remove the outermost electron from a gaseous atom, forming a positive ion. This energy is a measure of how tightly an electron is held to the nucleus. Factors like the attraction between the nucleus and electrons, electron repulsion, and effective nuclear charge all influence it. It decreases down a group and increases across a period in the periodic table.

- Periodic trends:
    - Across a period: It generally increases from left to right because the increasing nuclear charge pulls the electrons more tightly.

- Down a group: It decreases from top to bottom as the atomic radius increases, meaning the outermost electrons are farther from the nucleus and are shielded by more inner electrons.

Factors that influence ionization enthalpy

- Nuclear charge: A higher nuclear charge increases the force of attraction on electrons, requiring more energy to remove them.
- Atomic radius: A larger atomic radius means the valence electrons are farther from the nucleus, resulting in a weaker attraction and lower ionization enthalpy.
- Shielding effect: Inner-shell electrons "shield" the outer-shell electrons from the full force of the nuclear charge. A greater shielding effect results in a lower ionization enthalpy.

22. Normality

Definition: The number of equivalents of a solute per liter of solution.
Units: Equivalents/Liter (eq/L).
Use: Also called equivalent concentration, it is often used in acid-base and redox titrations.

Mole fraction

Definition: The ratio of the number of moles of a component to the total number of moles of all components in the solution.
Units: Unitless.

Parts per million (ppm)

Definition: A way to express very dilute concentrations, meaning one part of a substance per one million parts of the total solution.
Units: Commonly expressed as milligrams per liter (mg/L) for solutions or milligrams per kilogram (mg/kg).

Parts per billion (ppb)

Definition: A measure of concentration that signifies one part of a substance per one billion parts of the total mixture.
Units: Commonly expressed as micrograms per liter (µg/L) for solutions or micrograms per kilogram (µg/kg).